

A vintage computer monitor with a yellow frame is positioned on the left, displaying a green screen with white text. Below the monitor is a keyboard with a mix of black, white, and red keys. A snake with a yellow and black pattern is coiled on the desk to the right of the keyboard. The background is dark, and the overall scene is dimly lit.

Lecture 3: Digital Logic



Homework Assignment: Evil Hangman

Roadmap

- Review homework assignment
- Transistors
- Digital logic
- In-class exercise
- Bonus assignment

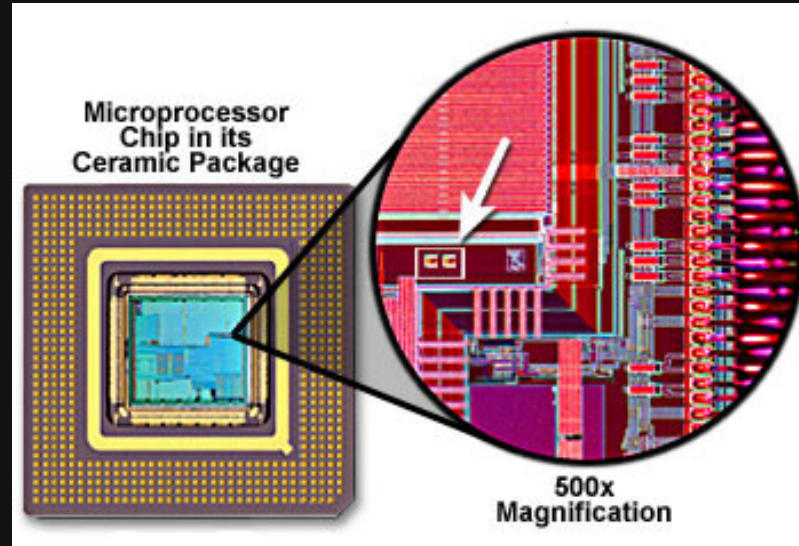


Transistors

- What is a transistor?
- Semiconductor device used to amplify or switch electrical signals
- Proposed in 1925
- First working device 1947
 - Nobel Prize (Physics) 1956
- Made from very pure silicon
- Can be used to make complex logic circuits



Transistors in Computer Chips





Moore's Law

Moore's Law: The number of transistors on microchips doubles every two years

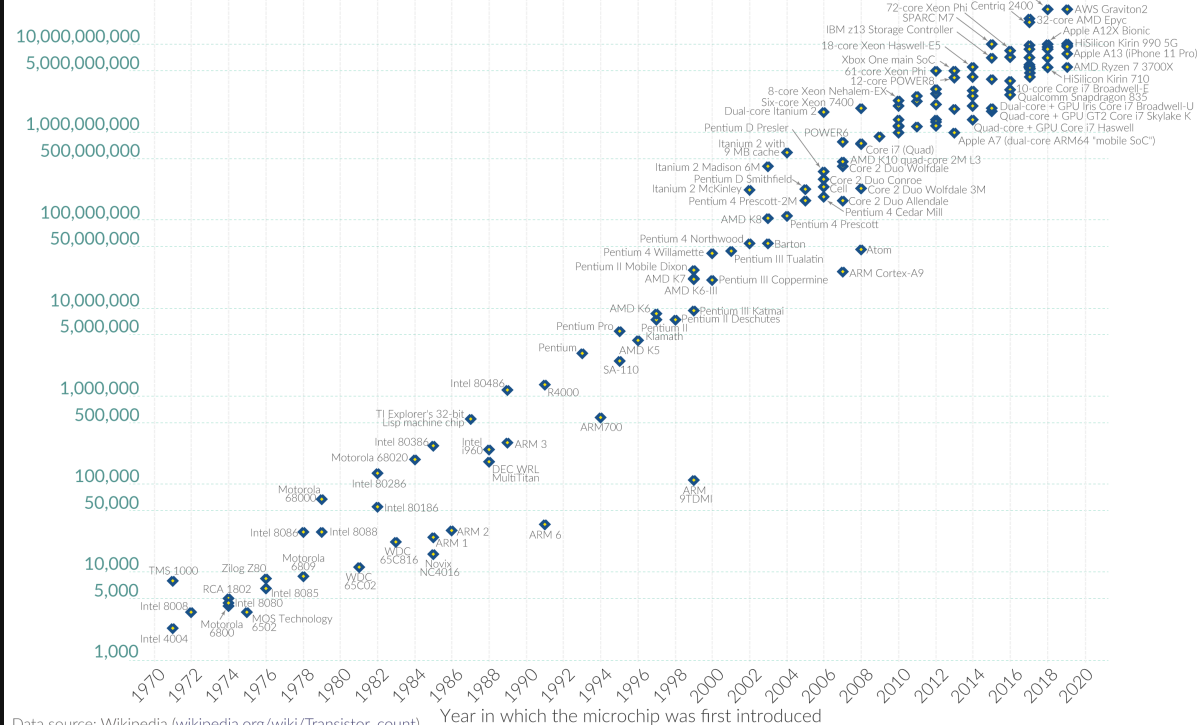
Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years.

This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count

50,000,000,000



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

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Logic Gates

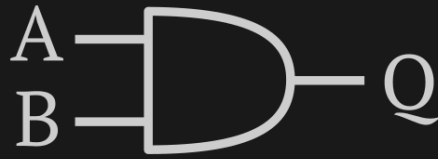
- Built from transistors acting as switches
- Implement Boolean logic
- Voltages indicate Boolean value
 - False: low logic level, usually 0 V
 - True: high logic level, usually 5 V, 3.3 V, 3 V, 1.8 V, 0.9 V
- Simulators:
 - <https://logic.ly/demo>
 - <https://circuitverse.org/simulator>

NOT Gate



A	Q
0	1
1	0

AND Gate



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1



OR Gate



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate



A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate



A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

XNOR Gate



A	B	Q
0	0	1
0	1	0
1	0	0
1	1	1



Exercise: 3-Way Vote

- Three toggle switches used for voting
- Light turns on if at least two switches are on
- Try it:
 - <https://logic.ly/demo>



3-Way Vote: Truth Tables

A	B	C	Q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

		A	
B	C	0	1
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1



3-Way Vote: Algebraic Notation

A	B	C	Q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

		A	
B	C	0	1
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

- NOT: $\overline{A}$
- AND: $A \cdot B$
- OR: $A + B$

$$Q = \overline{A} \cdot (B \cdot C) + A \cdot (B + C)$$

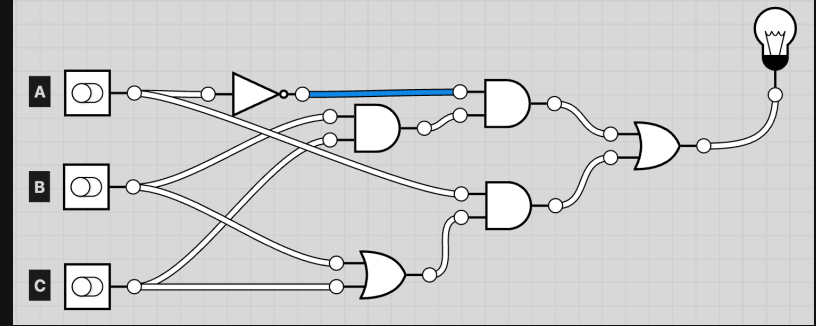


3-Way Vote: Solution

A	B	C	Q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

		A	
B	C	0	1
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

$$Q = \overline{A} \cdot (B \cdot C) + A \cdot (B + C)$$





Bonus Assignment: Rock, Paper, Scissors

- Bonus assignment:
 - Optional, no late penalties, can only help your grade
- Build a rock, paper, scissors game
 - Two players
 - Three switches per player
 - Two lights
 - If player 1 wins: light 1 on, light 2 off
 - If player 2 wins: light 1 off, light 2 on
 - Tips
 - Make truth tables for light 1
 - Write algebraic expression for light 1
 - Light 2 is opposite of light 1

